TBO VOYAGEHACK 4.0 · MULTI SUPPLIER OPTIMIZATION · TEAM ALPHAFIRE · PROTOTYPE

SupplyLens

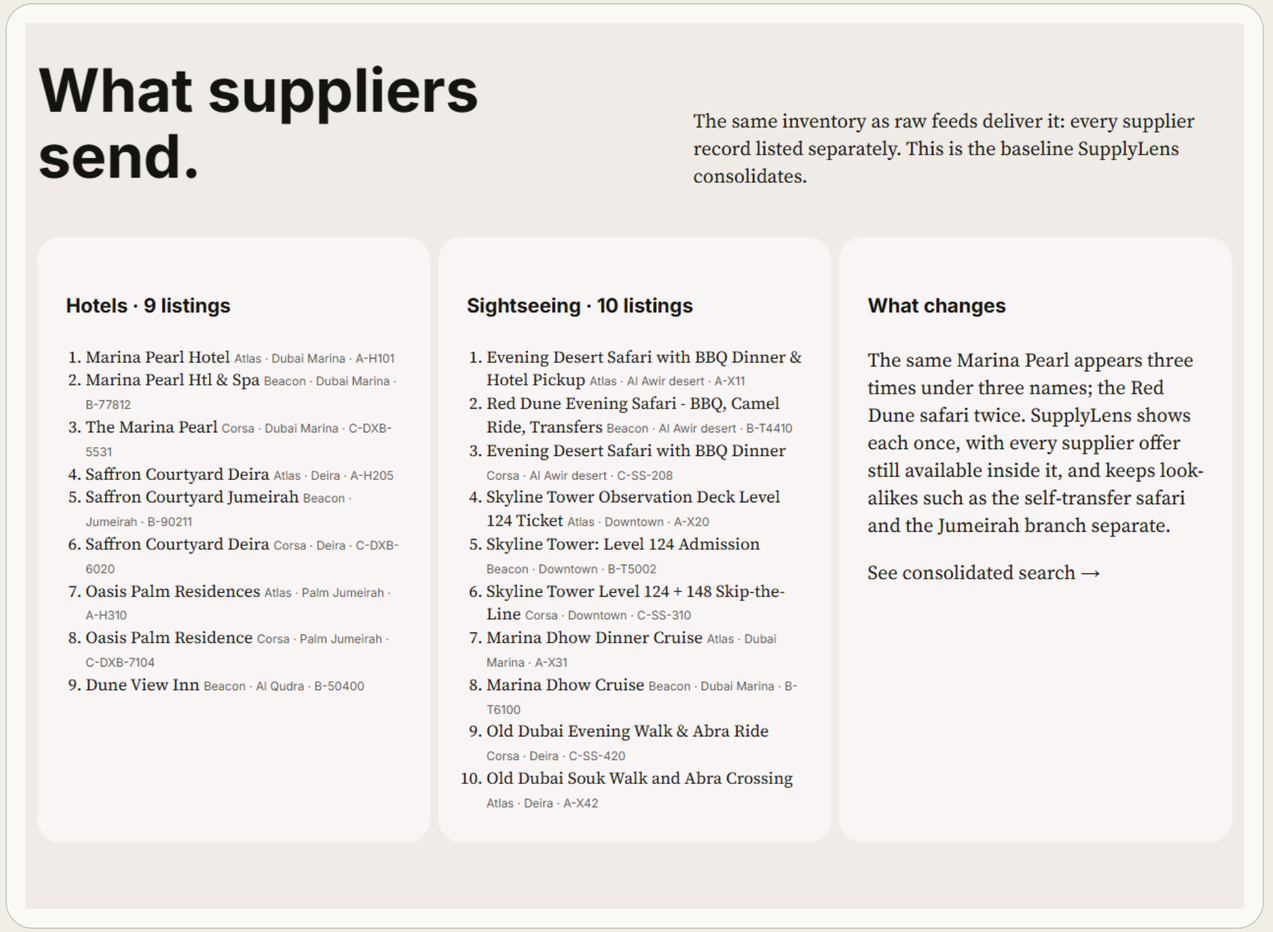
An explainable multi-supplier inventory layer for Hotels and Sightseeing: one trustworthy listing per product, distinct options kept, and the best eligible offer with its reasons.

`github.com/gurlivsingh/supplylens` · `supplylens-alphafire.onrender.com`

2 · THE PROBLEM

Feeds repeat products and hide differences.

- Marina Pearl arrives three times under three names
- A pickup safari and a self-transfer safari look identical by name
- The lowest headline price can exclude taxes, or have no refund at all
- Name-only deduplication causes 3 harmful merges per vertical on our benchmark



Raw supplier feed: 19 listings, synthetic

Synthetic data. All suppliers, prices and availability are fictional.

3 · APPROACH AND DIFFERENTIATION

Entity → variants → offers, with safeguards.

ENTITY

Marina Pearl Hotel

3 supplier records: “Marina Pearl Hotel” · “Marina Pearl Htl & Spa” · “The Marina Pearl”

→

VARIANTS

Room families

Deluxe King 32 m^2 (= 344 sqft) · Deluxe Twin · Superior King 45 m^2 · “Double or Twin, subject to availability” → only *possible*

→

OFFERS

Comparable only

AED 1,720 breakfast, free cancellation ✓
AED 1,650 non-refundable
~~AED 1,580~~ taxes excluded: not comparable

SIGHTSEEING

Red Dune evening safari

Same operator and itinerary from 2 suppliers → one product

≠

KEPT SEPARATE

Same safari, self-transfer

No hotel pickup, 5 h vs 6 h: a different product, never priced against the pickup version

Variant-safe consolidation
Identity never erases bed, meal, transfer, access or policy.

Best eligible, not lowest
Tax-exclusive, stale, unavailable or unknown-policy offers can't win.

Traceable and reversible
Reasons on every card; uncertain pairs wait for a reviewer, and every decision is logged.

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4 · ARCHITECTURE AND DATA GOVERNANCE

One small service, deterministic and auditable.

Ingest
supplier ids, raw records and timestamps kept

Normalize
abbreviations, sqft→m^2, phones, FX snapshot

Block
hotels ≤2 km · experiences by attraction

Decide
match / review / distinct with evidence and conflicts

Cluster
every pair must be compatible

Offers and rank
eligibility → labels → weighted, explained score

STACK

Python 3.12 · FastAPI · in-memory JSON fixture · single-page UI without a build step · Render blueprint. 19 automated tests. No secrets, no customer data, no bookings.

GOVERNANCE

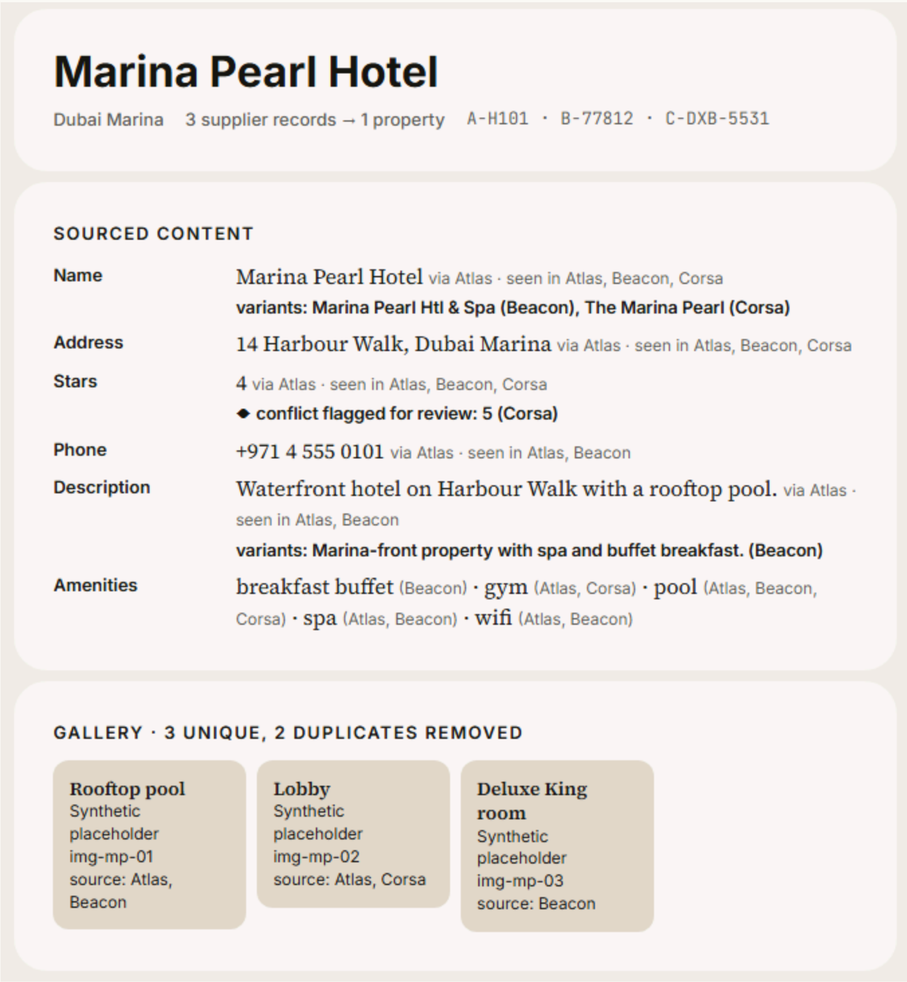
Field-level provenance; unknown stays unknown; synthetic data labelled on every screen; reviewer overrides in an append-only audit log; images are placeholders, deduplicated by hash.

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5 · HOTEL WORKFLOW

3 records → 1 property. 4 room families kept.

- Sourced fields; the star-rating conflict is flagged, not hidden
- 2 duplicate images removed, attribution kept
- 32 m^2 = 344 sqft → same family; King ≠ Twin
- “Subject to availability” bed → only *possible*



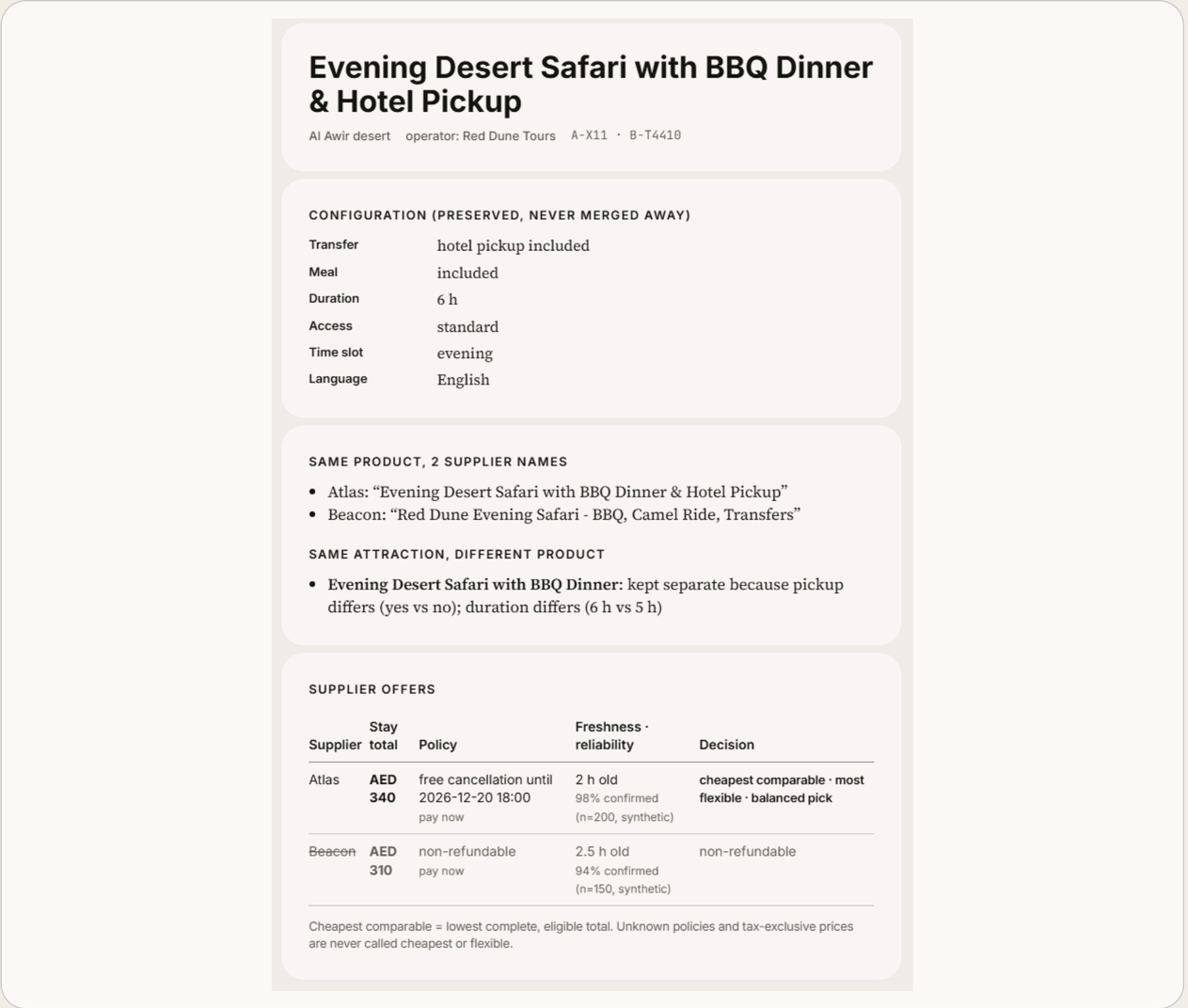
Marina Pearl: sourced content and gallery

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6 · SIGHTSEEING WORKFLOW

Same product merged. Look-alike kept apart.

- Red Dune safari: 2 suppliers, same operator and itinerary → 1 product
- Self-transfer version: pickup and duration differ → separate, with the reason shown
- Standard vs skip-the-line tower access: separate
- Missing operator → review queue, not a guess



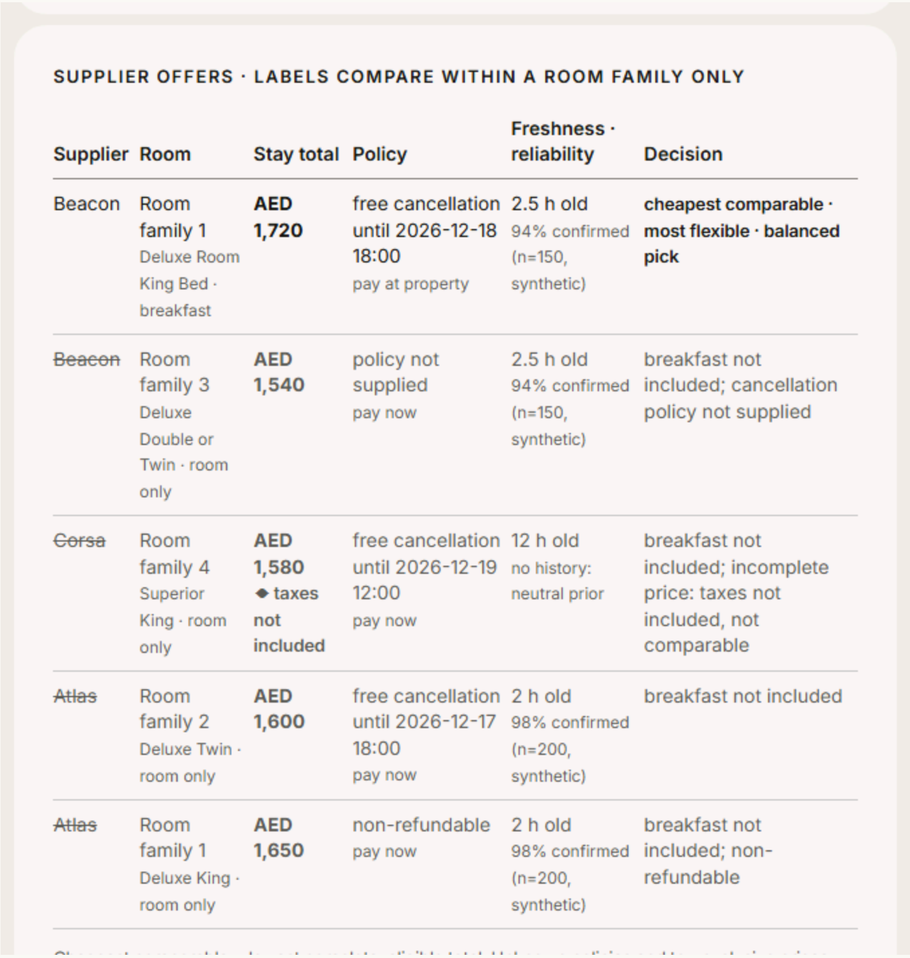
Red Dune evening safari: configuration preserved

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7 · SUPPLIER COMPARISON AND LISTING RANKING

Best eligible offer, then relevant order.

- Cheapest headline AED 1,580 loses: taxes excluded
- Unknown policy is never “free cancellation”
- Labels compare within one room family: cheapest comparable · most flexible · balanced pick
- Score = 0.40 relevance + relevance × (value, policy fit, freshness, reliability); new suppliers get a neutral prior



SUPPLIER OFFERS · LABELS COMPARE WITHIN A ROOM FAMILY ONLY

Supplier	Room	Stay total	Policy	Freshness · reliability	Decision
Beacon	Room family 1 Deluxe Room King Bed · breakfast	**AED 1,720**	free cancellation until 2026-12-18 18:00 pay at property	2.5 h old 94% confirmed (n=150, synthetic)	**cheapest comparable · most flexible · balanced pick**
~~Beacon~~	Room family 3 Deluxe Double or Twin · room only	**AED 1,540**	policy not supplied pay now	2.5 h old 94% confirmed (n=150, synthetic)	breakfast not included; cancellation policy not supplied
~~Corsa~~	Room family 4 Superior King · room only	**AED 1,580 ◆ taxes not included**	free cancellation until 2026-12-19 12:00 pay now	12 h old no history: neutral prior	breakfast not included; incomplete price: taxes not included, not comparable
~~Atlas~~	Room family 2 Deluxe Twin · room only	**AED 1,600**	free cancellation until 2026-12-17 18:00 pay now	2 h old 98% confirmed (n=200, synthetic)	breakfast not included
~~Atlas~~	Room family 1 Deluxe King · room only	**AED 1,650**	non-refundable pay now	2 h old 98% confirmed (n=200, synthetic)	breakfast not included; non-refundable

Offers: ineligible rows struck through, with the reason

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8 · EVALUATION (SYNTHETIC BENCHMARK, PER VERTICAL)

Zero harmful merges. Better ranking than both baselines.

IDENTITY MATCHING · 9 LABELLED PAIRS PER VERTICAL

	Hotels baseline	Hotels SupplyLens	Sightseeing baseline	Sightseeing SupplyLens
Auto-link precision	57%	**100%**	40%	**100%**
Harmful merges	3	**0**	3	**0**
Recall, auto / assisted	100 / 100%	100 / 100%	67 / 67%	67 / 100%
Sent to review	0	1	0	1

Baseline: name similarity ≥ 0.5. Candidate recall 100%. Offer rules: 0 violations in 43 recommendations. p95 search < 1 ms in-process.

LISTING RANKING · MEAN NDCG@K, HELD-OUT TEST INTENTS (5 PER VERTICAL)

	Raw feed	Cheapest eligible	SupplyLens
Hotels	0.62	0.96	**0.96**
Sightseeing	0.80	0.88	**0.97**

Validation intents (3 per vertical) were used for design: hotels 0.94, sightseeing 1.00. Small hand-labelled synthetic benchmark: exploratory, not production performance. Known failure: generic words like “tour” match every listing.

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9 · BUSINESS VALUE, FEASIBILITY AND INTEGRATION PATH

A low-risk path into TBO’s supplier pipeline.

VALUE MECHANISMS

- Fewer duplicate cards → faster, clearer choice
- Best eligible offer surfaced → price competitiveness
- Policy differences visible → fewer post-booking disputes
- Review queue focuses human effort on the uncertain few

FEASIBILITY

- Uses fields suppliers already send
- Deterministic, under 1 ms per search on the fixture
- No training data or GPU needed
- Runs as a sidecar service before the listing page

PROPOSED PILOT

1. Offline run on authorized TBO sample feeds, one city
2. Label 200 pairs; freeze thresholds
3. Shadow-rank against the live order
4. Controlled A/B on a slice of traffic; measure conversion there, not here

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10 · RECAP, LIMITATIONS AND NEXT STEP

Works today on synthetic data. Ready for a real-feed pilot.

WORKS IN THE SUBMITTED BUILD

- Hotel consolidation, enrichment, room mapping
- Sightseeing matching with preserved configurations
- Eligible-offer comparison and explained ranking
- Review queue with audit log; live evaluation tab; 19 tests

NOT YET VALIDATED

- Real supplier feeds and scale
- Lexical relevance only; no semantic or multilingual matching
- Reviewer decisions held in memory; no authentication
- No conversion claims without a live pilot

Our ask: authorized sample feeds for one destination, and a reviewer to label 200 pairs.

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